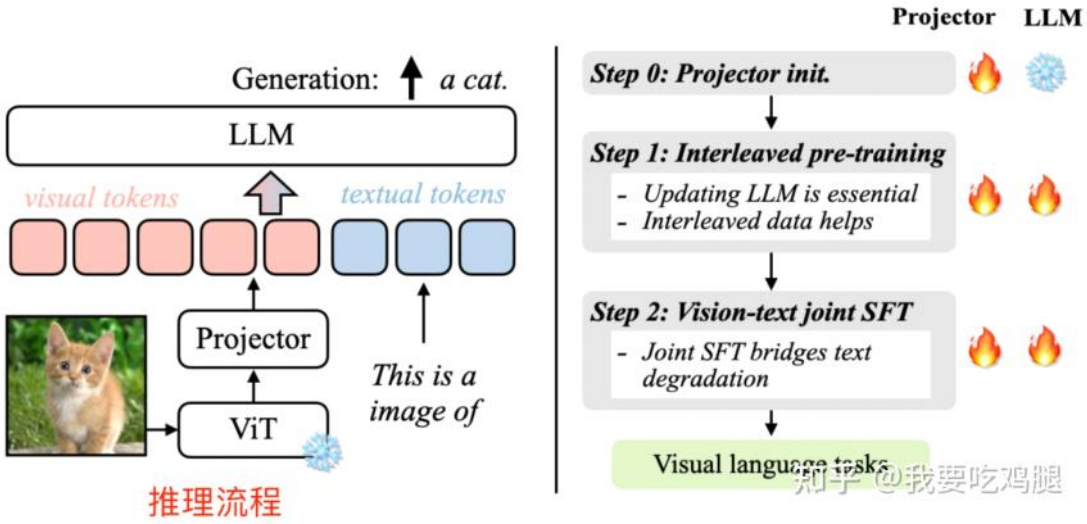
Spatial Reasoning

2026年4月5日 10:18

Overview :
https://mp.weixin.qq.com/s/FMnSoV6LuXW0Micp1BRGNw

总体共用架构



- AI **的“眼睛”（视觉编码器）**：负责最前端的感知。它的任务是将输入的像素世界——无论是静态图片还是动态视频，转化为机器能够理解的、蕴含丰富语义的数学表达（即特征向量）。
- AI **的“大脑”（大语言模型）**：负责最后端的认知。它拥有强大的语言理解、逻辑推理和内容生成能力，是整个系统的“思考中枢”。
- **灵魂之桥（连接器）**：负责最关键的沟通。它像一位精湛的“翻译官”，将“眼睛”看到的视觉信息，精准地翻译成“大脑”

能够听懂的“语言”，实现两大模态的无缝对接。

1. ReAct (2023 ICLR) Reasoning Traces + Task-specific Reasoning

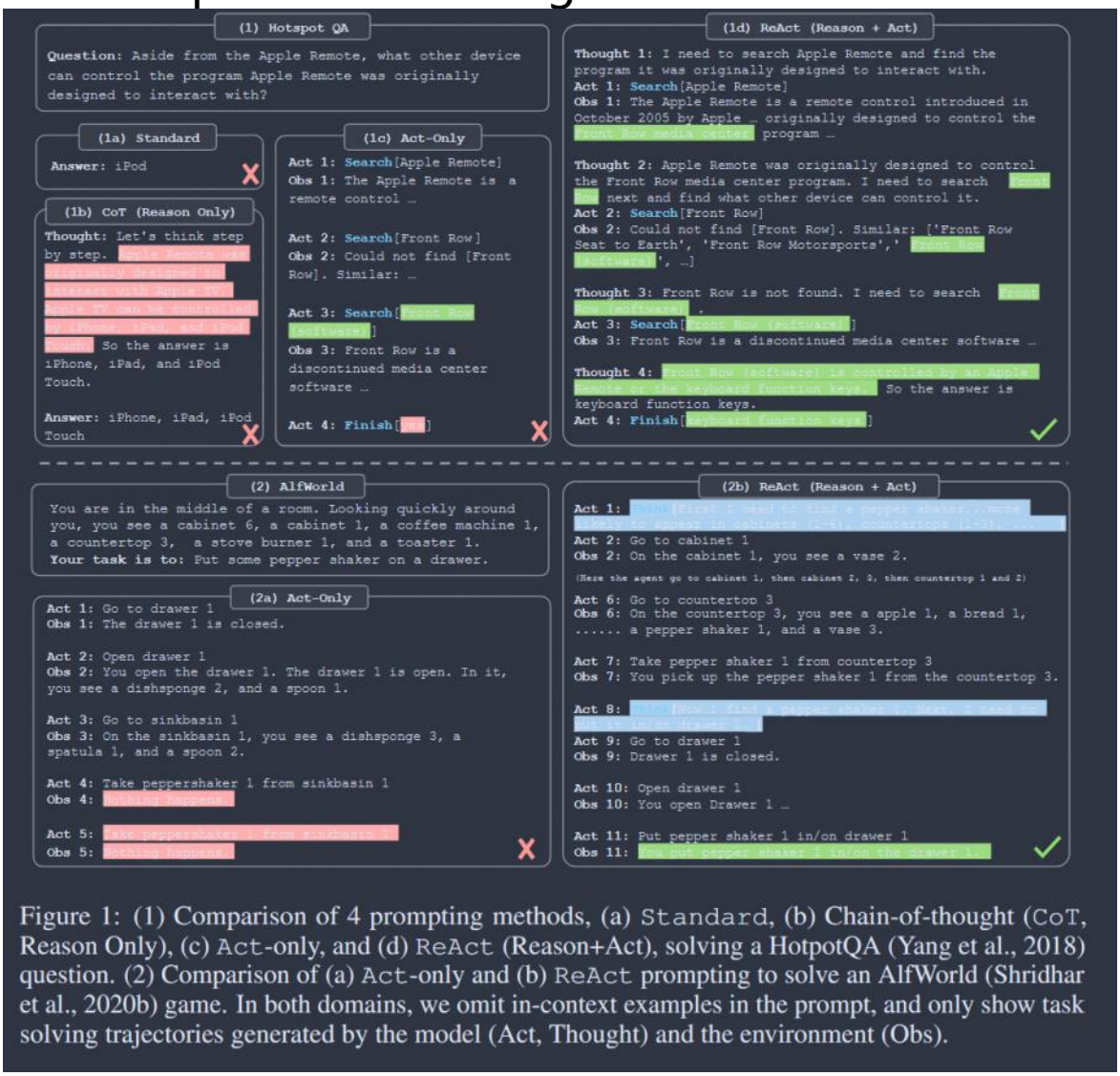


Figure 1: (1) Comparison of 4 prompting methods, (a) `Standard`, (b) Chain-of-thought (`CoT`, Reason Only), (c) `Act`-only, and (d) `ReAct` (Reason+Act), solving a HotpotQA (Yang et al., 2018) question. (2) Comparison of (a) `Act`-only and (b) `ReAct` prompting to solve an AlfWorld (Shridhar et al., 2020b) game. In both domains, we omit in-context examples in the prompt, and only show task solving trajectories generated by the model (Act, Thought) and the environment (Obs).

Idea:

The idea of `ReAct` is simple: we augment the agent's action space to $\hat{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$, where $\mathcal{L}$ is the space of language. An action $\hat{a}_t \in \mathcal{L}$ in the language space, which we will refer to as a *thought* or a *reasoning trace*, does not affect the external environment, thus leading to no observation feedback. Instead, a thought $\hat{a}_t$ aims to compose useful information by reasoning over the current context c_t, and update the context $c_{t+1} = (c_t, \hat{a}_t)$ to support future reasoning or acting. As shown in Figure 1, there could be various types of useful thoughts, e.g. decomposing task goals and create action plans (2b, Act 1; 1d, Thought 1), injecting commonsense knowledge relevant to task solving (2b, Act 1), extracting important parts from observations (1d, Thought2, 4), track progress and transit action plans (2b, Act 8), handle exceptions and adjust action plans (1d, Thought 3), and so on.

动作（action）序列 $\hat{A} = A \cup \mathcal{L}$ ，引入语言模型，相当于引入人类的思考来决策下一步动作。

针对HotpotQa, Fever数据集，需要知识增强调用wikipedia API:

Action包括：

- **search**[entity]：返回页面的前5句话，若不存在，则返回最相似的top-5实体
- **lookup**[string]：查找页面中包含字符串的句子，或者在浏览器中crtl+F
- **finish**[answer]：返回当前任务的答案。

Exp:

结果：

- ReAct + CoT-SC 结合的效果最好
  - ReAct -> CoT-SC：当ReAct在7（HotpotQA）、5（Fever）没有给出答案时，改用COT-SC
  - CoT-SC -> ReAct：当采样得到的n个采样结果中，没有一个输出又超过n/2投票，结果不可信，改用ReAct
- 微调之后ReAct效果进一步提升。

Prompt Method[4]	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC → ReAct	34.2	**64.6**
ReAct → CoT-SC	**35.1**	62.0
Supervised SoTA[4]	67.5	89.5

Table 1: PaLM-540B prompting results on HotpotQA and Fever.

[4] HotpotQA EM is 27.1, 28.9, 33.8 for Standard, CoT,

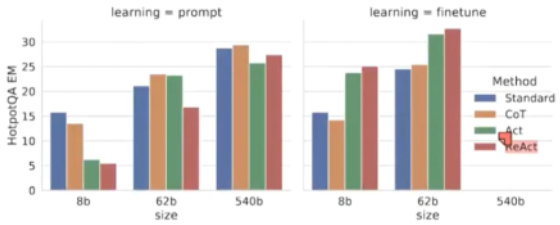


Figure 3: Scaling results for prompting and finetuning on HotPotQA with ReAct (ours) and baselines.

2. Spatial VLM (2024 CVPR) - VLM data synthesis and backbone model

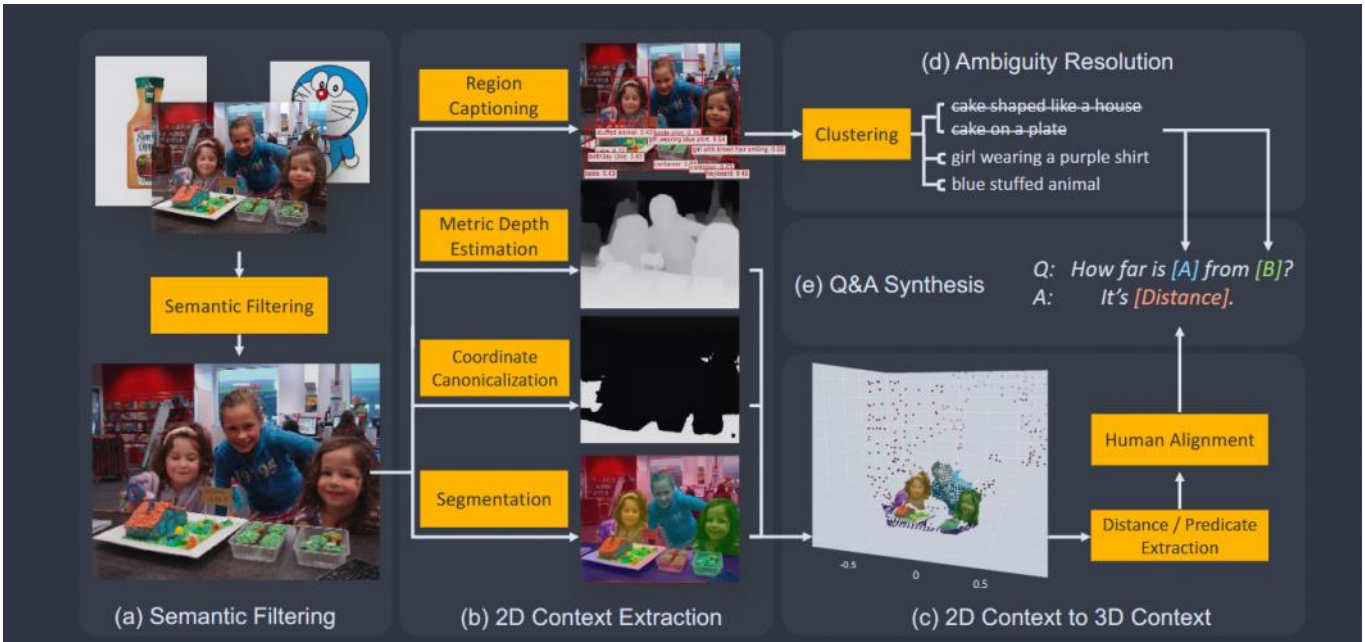


Figure 2. **An overview of our data synthesis pipeline.** (a) We use CLIP to filter noisy internet images and only keep scene-level photos. (b) We apply pre-trained expert models on internet-scale images so that we get object-centric segmentation, depth and caption. (c) We lift the 2D image into 3D point clouds, which can be parsed by shape analysis rules to extract useful properties like 3D bounding box. (d) We avoid asking ambiguous questions by clustering object captions using CLIP similarity score (e) We synthesize millions of spatial question and answers from object captions and extracted properties.

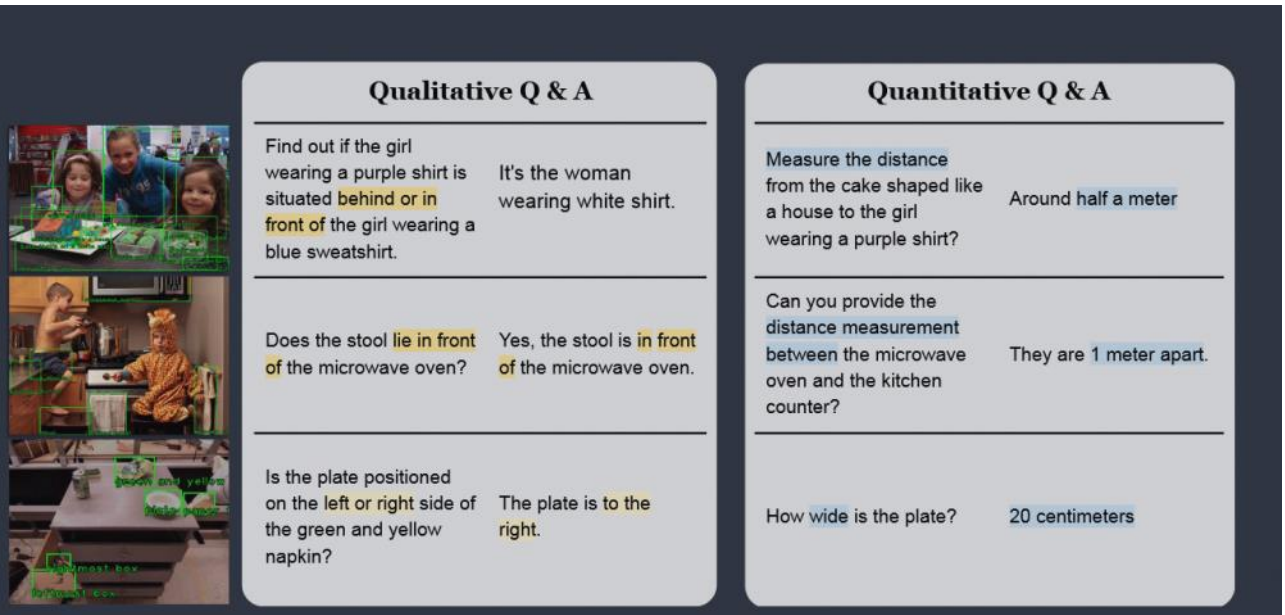


Figure 3. **Example data entries from the synthetic dataset**. Given the output of vision expert models, we follow a set of question generation template to generate both quantitative and qualitative question-answer pairs to highlight the diversity of the dataset. The spatial concepts are highlighted in blue. Such visual question-answer pairs can be easily mixed together with other captioning or question answering datasets and use the same training objectives.

Model structure

Direct Spatial Reasoning is defined as following, a Vision-Language Model takes as input an image $\mathcal{I}$ and a query $\mathcal{Q}$ of a spatial task, and output an answer $\mathcal{A}$, in the format of a text string, without using external tools or interacting with other large models. We adopt the same architecture and training procedure of PaLM-E [18] except replacing PaLM [14] backbone with PaLM 2-S [3], a smaller variant. We then train our model using a mixture of the original PaLM-E dataset and our dataset, with 5% of tokens dedicated to spatial reasoning tasks. Similar to PaLM-E, our method has the ability to perform VQA as well as basic embodied planning when combined. The key difference is that it can answer spatial reasoning questions about both binary predicates and quantitative estimations.

Pre:

LLaVA **系列**。它所奉行的核心哲学可以概括为：“**保持模型核心的极简性，通过在输入端进行巧妙的工程设计，来‘欺骗’模型，使其能消化远超其原始训练分辨率的信息。**”。

LLaVA 的开发者们没有选择去重构模型内部复杂的融合机制，而是将智慧聚焦于数据预处理阶段，提出了一套名为 AnyRes（Any Resolution）的解决方案。顾名思义，它的目标就是让模型能够处理“任意分辨率”的图像。

LLaVA-1.5-HD：**“全局+局部”的巧妙协同**

AnyRes 的思想在 LLaVA-1.5-HD 版本中得到了经典的体现。其策略的核心是“两手抓，两手都要硬”，同时兼顾图像的全局上下文和局部细节。

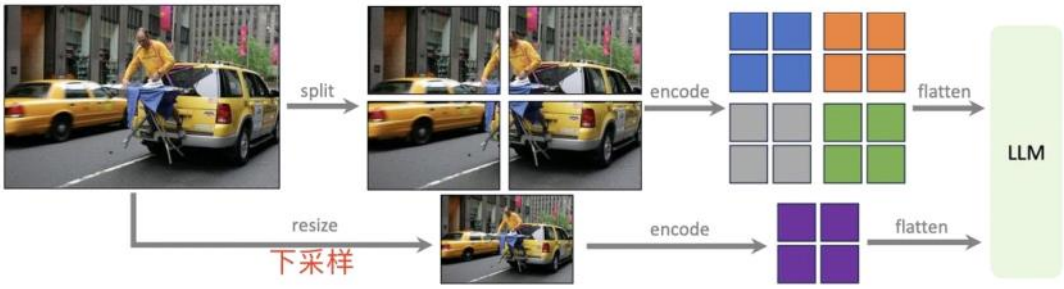


Figure 2. **LLaVA-1.5-HD.** Scaling LLaVA-1.5 to higher resolutions by splitting the image into grids and encoding them independently. This allows the model to scale to any resolution, without performing positional embedding interpolation for ViTs. We additionally concatenate the feature of a downsampled image to provide the LLM with a global context.

如上图所示，当 LLaVA-1.5-HD 接收到一张高分辨率图像时，它会兵分两路进行处理：

1. **“局部细节”处理路径（上路）：**
   - 模型首先将原始的高清大图（High Resolution Image）**分割**（split）成多个较小的、但依然保持高清晰度的图块。这些图块的尺寸通常与视觉编码器（如 CLIP-ViT）原始训练时能够处理的最大分辨率（例如 336x336）相匹配。
   - 然后，视觉编码器会**独立地**（independently）对这些高清图块进行**编码**（encode）。由于每个图块都保持了足够的分辨率，图像中的微小文字、物体细节等关键信息得以被完整地保留下

来。

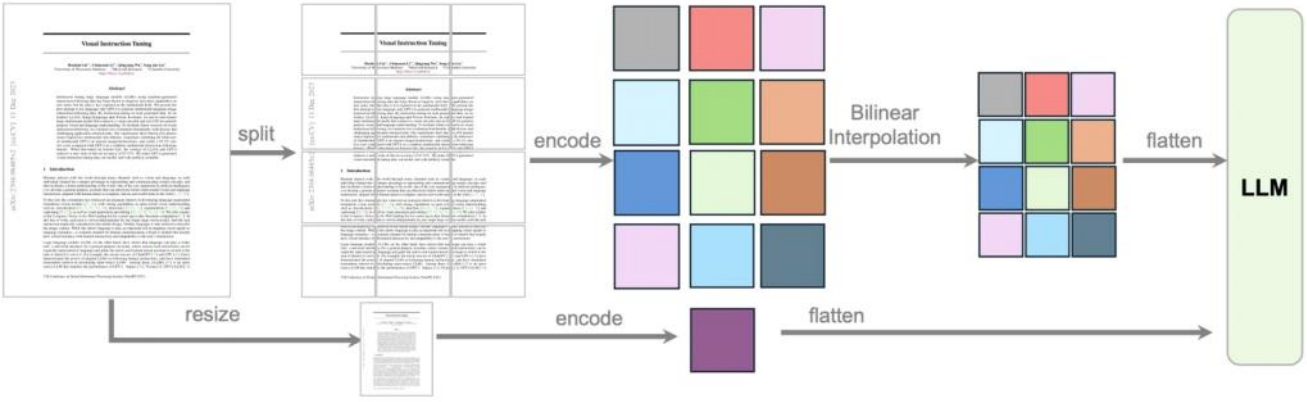
2. **“全局上下文”处理路径（下路）**：
    - 与此同时，模型会将原始的高清大图进行一次**降采样**（resize / downsample），将其缩放到一个标准尺寸，形成一张低分辨率的全局概览图。
    - 这张全局图随后也被送入视觉编码器进行**编码**（encode）。虽然这个过程会损失大量细节，但它为模型提供了关于整张图片“讲的是什么”的宝贵**全局上下文**（global context）。
3. **最终融合：**
    - 最后，来自“局部细节”路径的所有高清图块特征，与来自“全局上下文”路径的单一全局特征，会被**拼接**（concatenate）在一起，形成一个更长的视觉序列，共同送入 LLM。

通过这种巧妙的协同，LLM 最终得到的输入，既包含了来自各个局部的高保真细节，又具备了对整张图片布局和主题的宏观理解，从而实现了对高分辨率图像的精准解析。

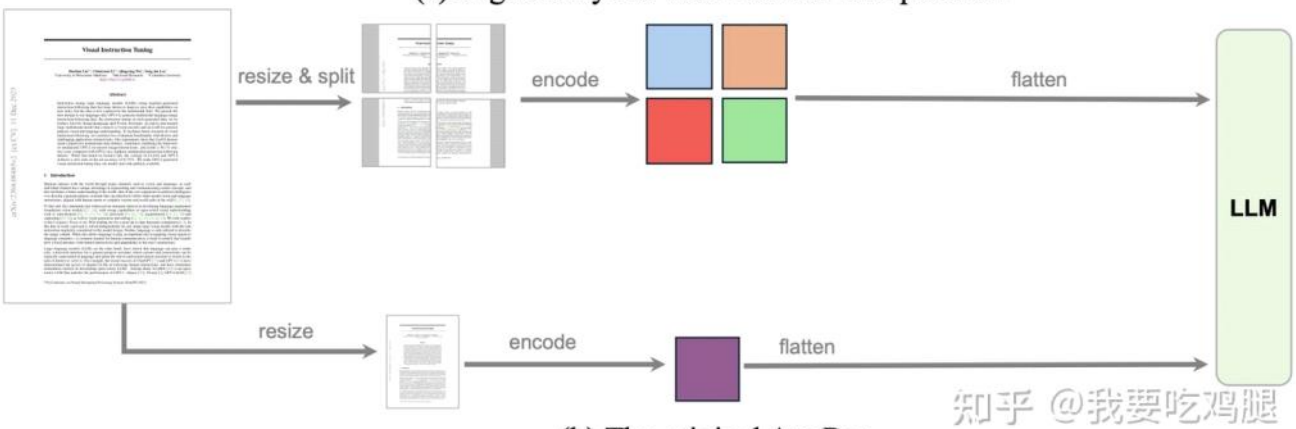
LLaVA-OneVision **的进阶：为文档而生的**

高级 AnyRes

随着任务的复杂化，尤其是在处理 PDF 文档、学术论文这类信息密度极高的场景时，简单的特征拼接已经不足以完美地保留图块之间的空间关系。为此，LLaVA-OneVision 提出了更先进的 Higher AnyRes 策略。



(a) Higher AnyRes with Bilinear Interpolation



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(b) The original AnyRes

对比上图中的（a）和（b）两种方案，我们可以清晰地看到其核心区别：

- **原始** AnyRes（The original AnyRes）：如图（b）所示，它在编码各个图块后，直接将得到的特征向量**展平**（flatten）并拼接。这种方式虽然简单，但可能会丢失图块之间精细的二维布局信息。

- **高级** AnyRes (Higher AnyRes with Bilinear Interpolation)：如图 (a) 所示，它在编码完各个高清图块后，增加了一个关键步骤——**双线性插值** (Bilinear Interpolation)。它不再是将特征向量粗暴地拼接，而是尝试在特征空间中，通过插值的方式将这些离散的特征图块“缝合”起来，重新构建一个更高分辨率的**特征图** (Feature Map)。然后再将这个融合了空间信息的完整特征图展平送入 LLM。这种方式能更好地保留文档的布局、表格的结构等至关重要的空间关系，极大地提升了模型在文档视觉问答 (DocVQA) 等任务上的表现。

小结一下，LLaVA 系列的演进之路，充分展现了其“扩展输入边界”的设计哲学。它通过不断迭代 AnyRes 这一核心前端技术，在不改动模型核心架构的前提下，聪明地解决了高分辨率视觉信息的处理难题，证明了优秀的工程设计与数据处理策略，同样是推动模型能力飞跃的关键力量。

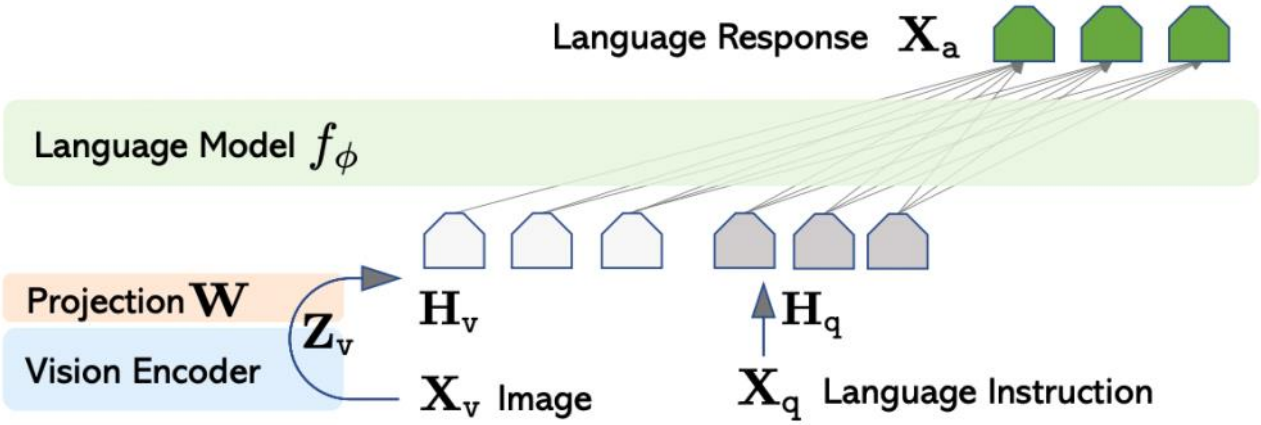


Figure 1: LLaVA network architecture.

- **Vision Encoder:** pre-trained CLIP visual encoder ViT-L/14
- **LLM:** Vicuna
- **Projection:** a simple linear layer to connect image features into the word embedding space。不同于Flamingo的gated cross-attention，和BLIP-2的Q-former，LLaVA的映射层比较简单，好处是可以方便迭代以数据为中心的实验

训练方法：

- **阶段一：特征对齐的预训练（**Pre-training for Feature Alignment**）。**
  - 冻结视觉编码器和LLM，只训练Projection层。
  - 从CC3M中过滤了595K的图文对。这些图文对被改造为指令跟随数据（instruction-following data）
  - 每个样本是一个单轮对话。

- 预测结果是图像的原始描述文本。
- 通过这一步，视觉编码器输出的图像特征可以和预训练的LLM的词向量对齐。这个阶段可以理解为**为冻结的**LLM **训练一个兼容的**visual tokenizer**。**
- 在CC-595K过滤后的子数据集上训练，epoch数量为1，学习率2e-3，batch_size为128。

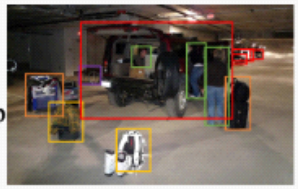
- **阶段二：端到端微调（**Fine-tuning End-to-End**）。**
  - 冻结视觉编码器，训练Projection层和LLM。
  - 两个实验设置
    - 多模态对话机器人：利用158K语言-图像指令跟随数据对Chatbot进行了微调开发。在这三种类型的回答中，对话型是多轮交互式的，而其他两种则是单轮交互式。在训练过程中，这些样本被均匀地采样使用。
    - ScienceQA数据集：ScienceQA [34] 包含了21,000个多模态多项选择题，这些问题涵盖了广泛的领域多样性，涉及3个学科、26个主

题、127个类别和379项技能。这个基准数据集被划分为训练、验证和测试三个部分，分别包含12,726个、4,241个和4,241个样本。

- 在提出的LLaVA-Instruct-158K数据集上训练，epoch数量为3，学习率2e-5，batch_size为32。

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], person: [0.63, 0.222, 0.686, 0.516], person: [0.444, 0.233, 0.487, 0.34], backpack: [0.384, 0.696, 0.485, 0.914], backpack: [0.755, 0.413, 0.846, 0.692], suitcase: [0.758, 0.413, 0.845, 0.69], suitcase: [0.1, 0.497, 0.173, 0.579], bicycle: [0.282, 0.363, 0.327, 0.442], car: [0.786, 0.25, 0.848, 0.322], car: [0.783, 0.27, 0.827, 0.335], car: [0.86, 0.254, 0.891, 0.3], car: [0.261, 0.101, 0.787, 0.626]

Response type 1: conversation

Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV).
Question: Where is the vehicle parked?
Answer: The vehicle is parked in an underground parking area, likely in a public garage.
Question: What are the people in the image doing?
Answer: The people are trying to fit all of their luggage into the SUV, likely preparing for a trip.

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip.
In the parking area, various luggage items are scattered around the vehicle. There are two backpacks, one located near the left rear wheel and the other closer to the right side of the vehicle. Additionally, there are two suitcases, one on the right side of the car and another further away near the center of the parking area. A bicycle can also be seen on the left side of the vehicle.
Other cars are parked around the main SUV, with one car positioned behind it and slightly to the left, another behind and slightly to the right, and the third car further behind on the right side.

Response type 3: complex reasoning

Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings to accommodate. They might have to strategize and arrange the luggage efficiently to ensure that everything fits properly into the vehicle. Additionally, they need to consider the comfort of the passengers and visibility while driving, so the placement of the luggage must not obstruct the driver's view or make the passengers uncomfortable during the trip.

3. LLaVA CoT - Step by step Reasoning

3.1.1. Reasoning Stages

Our proposed model, LLaVA-CoT, decomposes the answer generation process into four structured reasoning stages:

- **Summary Stage.** In this initial phase, LLaVA-CoT provides a high-level summary interpretation of the question, outlining the primary aspects of the problem it intends to address.
- **Caption Stage.** LLaVA-CoT offers a concise overview of the visual elements relevant to the question, helping to understand multimodal input.
- **Reasoning Stage.** Building on the initial summary, LLaVA-CoT conducts structured, logical reasoning to derive a preliminary answer.
- **Conclusion Stage.** In this final stage, LLaVA-CoT synthesizes an answer based on the preceding reasoning. Here, the output from the conclusion stage is the direct response provided to the user, while the prior three stages are internal "hidden stages" representing LLaVA-CoT's reasoning process. The output at this stage adapts to the user's requirements: for instance, if the user requests a brief answer, the conclusion will be concise; if detailed explanations are desired, the conclusion provides a thorough, comprehensive response.

Inference:

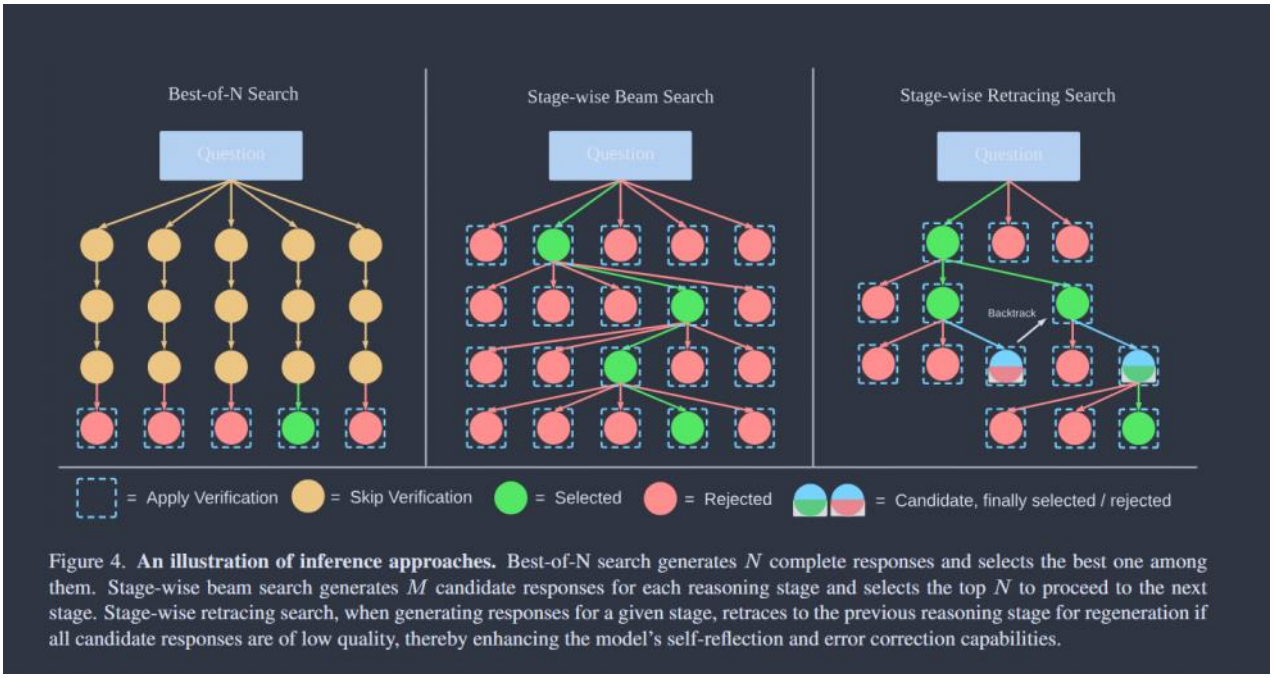


Figure 4. **An illustration of inference approaches.** Best-of-N search generates N complete responses and selects the best one among them. Stage-wise beam search generates M candidate responses for each reasoning stage and selects the top N to proceed to the next stage. Stage-wise retracing search, when generating responses for a given stage, retraces to the previous reasoning stage for regeneration if all candidate responses are of low quality, thereby enhancing the model's self-reflection and error correction capabilities.

4. NavCoT - 解决离线使用下VLN task与LLM训练语料的差距

parameter-efficient in-domain training to enable self-guided navigational decision

Introducing world model into CoT mechanism

关于两种不同的LLM在VLN中应用:

B. Large Language Models (LLMs)

Applying LLMs to non-linguistic embodied tasks has received more and more attention recently. We roughly divide these methods into two categories: the first one is to employ closed-source LLMs such as GPT-4 [16] in an offline way [9], [14], [39], which may suffer from poor scalability and severe domain gap. The second one is to introduce smaller open-source LLMs which can be deployed and trained locally [42]. Our method lies in the latter, and we adopt two open-source LLMs, LLaMA-Adapter [20] and LLaMA 2 [8], as the navigation backbones.

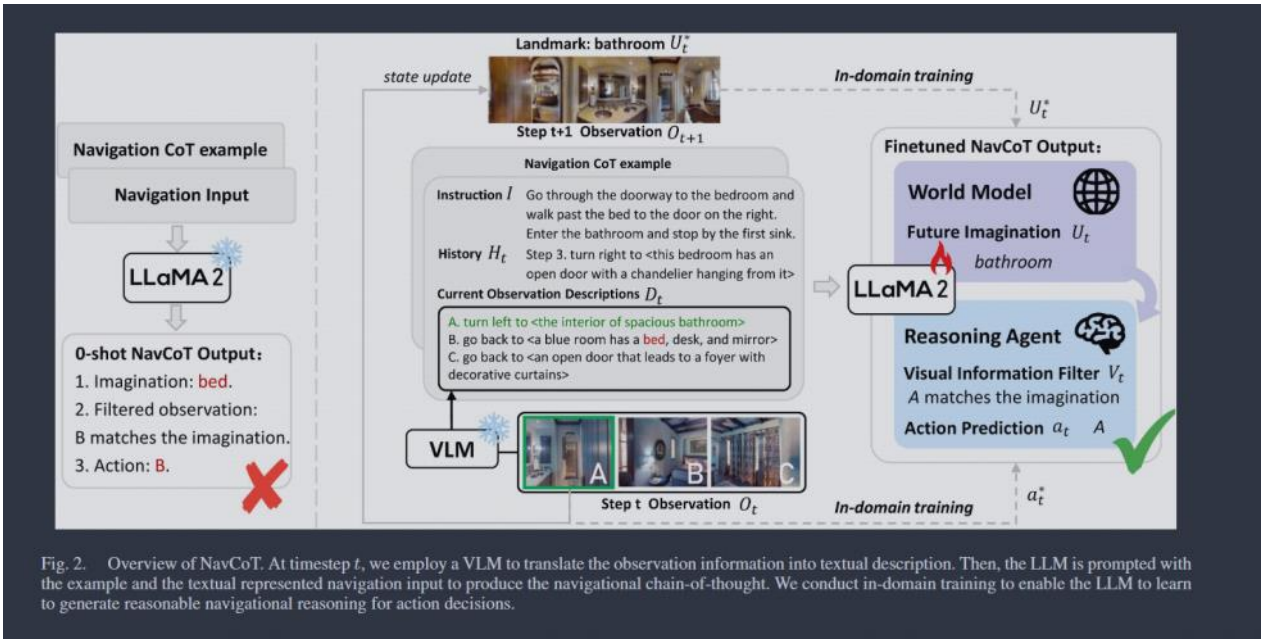


Fig. 2. Overview of NavCoT. At timestep t, we employ a VLM to translate the observation information into textual description. Then, the LLM is prompted with the example and the textual represented navigation input to produce the navigational chain-of-thought. We conduct in-domain training to enable the LLM to learn to generate reasonable navigational reasoning for action decisions.

论文将 NavCoT 的运行机制拆解为以下几个关键步骤：

- A. 视觉到文本系统 (Vision-to-Text System)：利用图像描述模型（BLIP）将视觉信息转化为文本，并结合方位信息（如"左转"、"向上"），将环境输入转换为 LLM 能理解的纯文本描述。 +1
- B. 导航思维链提示 (Navigational Chain-of-Thought Prompt)：将推理过程解耦为三个严格步骤：
  - **未来想象 (FI)**：LLM 根据指令预测接下来可能遇到的物体或场景。 +1
  - **视觉信息过滤 (VIF)**：从当前冗余的观测视图中，挑选出最符合刚才"想象"的视图。 +1
  - **动作预测 (AP)**：总结上述两步推理，输出最终的动作决策。
- C. 推理真值收集 (Reasoning Ground-Truth Collection)：为了训练模型，作者使用 LLM 提取指令中的地标，并用 CLIP 计算图像与地标的相似度，从而自动构建了高质量的"想象"和"过滤"的真实标签 (Ground-Truth)。 +1
- D. 域内 CoT 训练 (In-Domain Chain-of-Thought Training)：采用任务分解的预训练（将 FI、VIF、AP 作为独立任务）和基于模仿学习的微调，让 LLM 真正学会按预期格式生成解耦的导航推理。

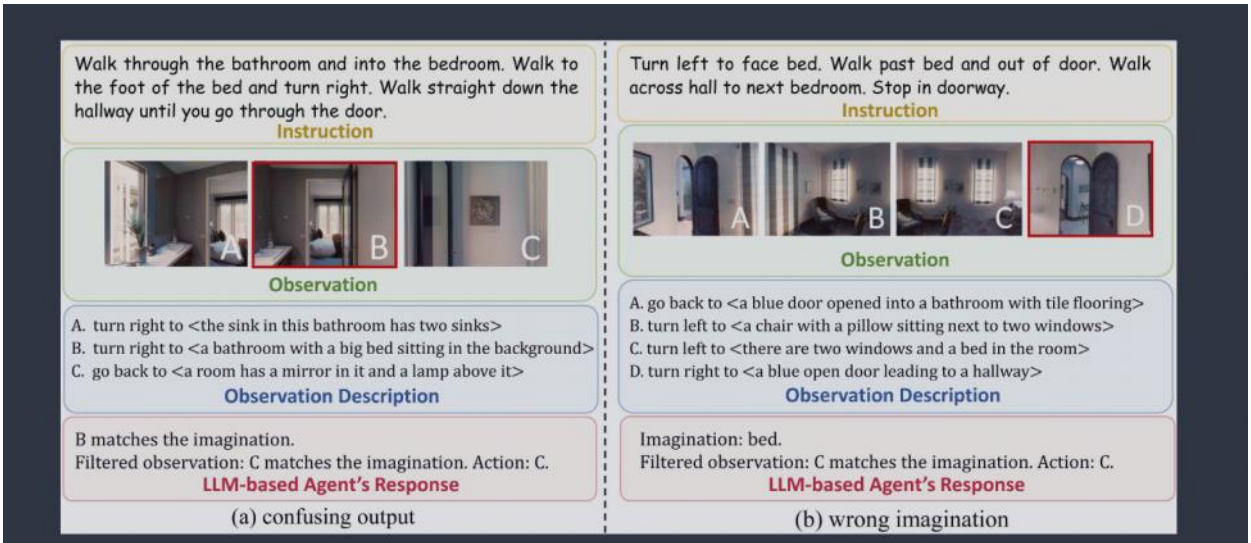


Fig. 3. Failure cases of LLM output in the zero-shot manner. The ground-truth actions are denoted by red boxes.

5. Geometrically-Constrained Agent: 2025 - VLM缺乏高精度几何计算能力和空间朝向判断

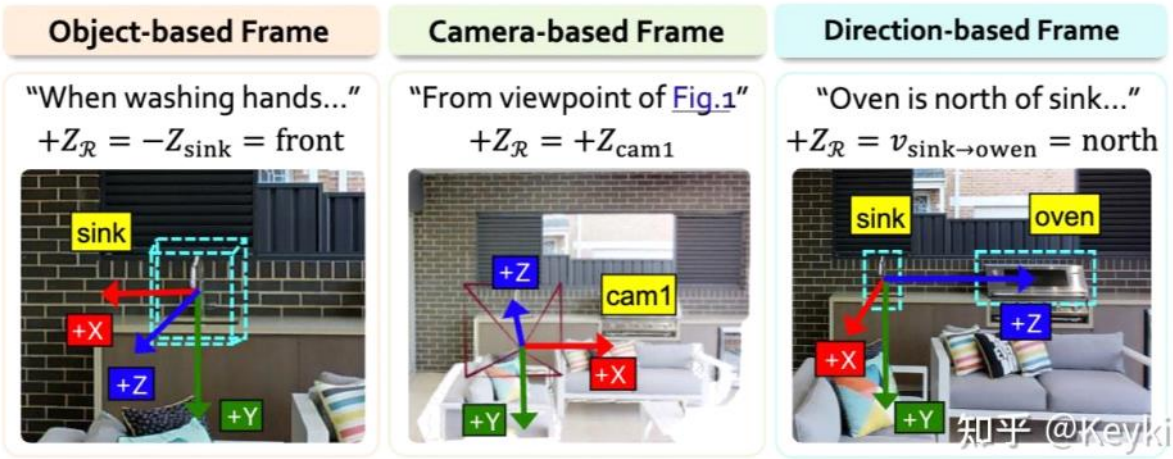
形式化的任务约束

两阶段推理：

1. 任务分解。把模糊的自然语言任务转化为（1）参考系；（2）明确要解答的「几何问题」本身，例如具体的距离值、角度或方位。
   a. 三类参考系：
   - **基于对象的参考系（**Object-based Frame**）**：例如“洗手时”，参考系的 +Z 轴必须与水槽语义上的正前方（-Z_{sink}）对齐。
   - **基于相机的参考系（**Camera-based

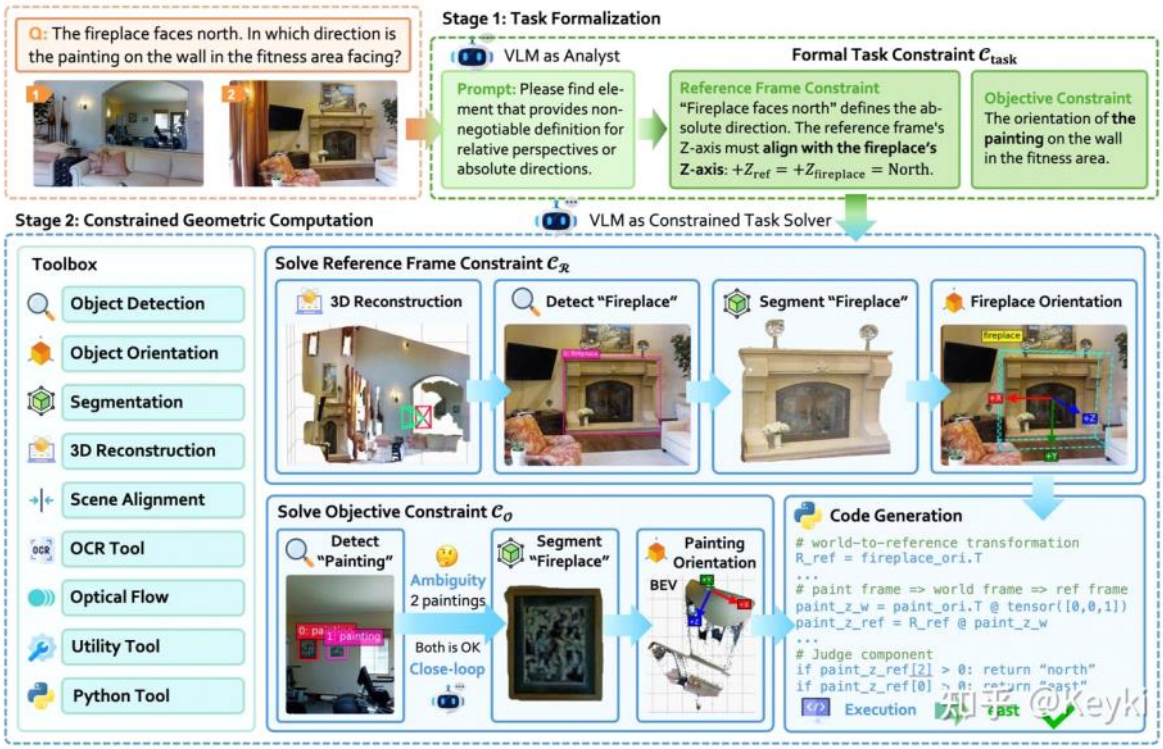
Frame）：例如“从图1的视角看”，参考系就是相机的坐标系（+Z_{cam1}）。

- **基于方向的参考系（Direction-based Frame）**：例如“烤箱在水槽的北边”，参考系由两个物体中心点构成的向量来定义



b. 关键目标。在确定的参考系下，明确需要测量的具体目标，比如“椅子相对于烤箱的位置关系。

c. 执行。在约束边界下，调用外部工具进行计算。



例子：

- Stage 1：任务分解。参考系为壁炉的朝向。关键目标为健身房的墙上的画。
- Stage 2：VLM 生成工具调用计划。
  - 为了找到壁炉，需要（1）调用3D重建工具获取场景；（2）调用目标检测工具找到壁炉；（3）调用分割和姿态估计工具找到壁炉的3D朝向。
  - 为了找到健身房墙上的画，需要（1）目标检测工具找到墙上的画；（2）结合分割的图像和文本，让VLM做歧义消除，解决工具调用后输出内容包含噪声的问题。
  - 调用工具拿到所有内容后，VLM 调用

python 进行最终计算。特别的，计算公式是预先定义好的，会根据当前问题去调用预定义的公式（如局部到全局的坐标转换公式、旋转矩阵转换等），以避免VLM产生数据公式“幻觉”。

工具：

- **几何与感知工具**：使用 VGGT 进行全局统一的高保真 3D 重建，利用 MoGe-2 获取具备物理度量尺度（Metric Scale）的深度图，结合 Grounding DINO（开放词汇检测）、SAM-2（实例分割）和 Orient Anything（3D 朝向估计）来参数化视觉世界 。
- **计算与实用工具**：除了核心的 Python 沙盒，还包括 project_box_to_3d_points（将 2D 像素框转换为 3D 点云）和光流分析等功能，实现 2D 语义到 3D 几何的无缝映射 。

6. 场景图与LLM对齐机制：

3DGraphLLM:Combining Semantic Graphs and Large Language Models for 3D Scene Understanding (ICCV 2025) ——加强语义信息关系的输入

https://github.com/cognitiveaisystems/3dgraphllm

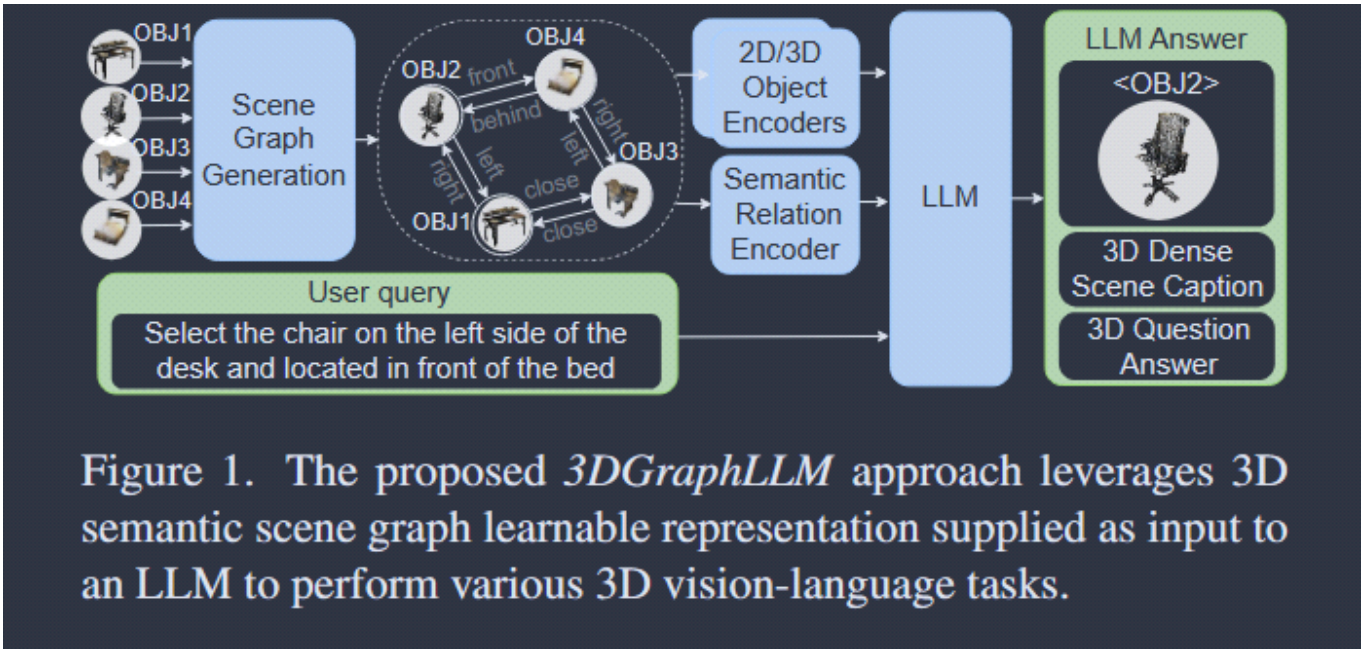


Figure 1. The proposed *3DGraphLLM* approach leverages 3D semantic scene graph learnable representation supplied as input to an LLM to perform various 3D vision-language tasks.

Problem Setup: 3DScene Graph + Scene Representation for LLM

In this paper, we introduce 3DGraphLLM, a novel learnable representation of a 3D scene graph designed for use as input to an LLM (see Fig. 1). The representation consists of a list of learnable embeddings for scene objects, where each object is modeled as a local subgraph that includes the object itself and its nearest neighbors. These subgraphs are provided to the LLM as a sequence of triplets *(object1, relation, object2)*. Semantic relations are encoded using features derived from the semantic edges of the scene graph, generated by state-of-the-art methods such as VL-SAT [52].

Pre-trained encoders for 2D images, 3D point clouds and point clouds semantic relationships + LLM

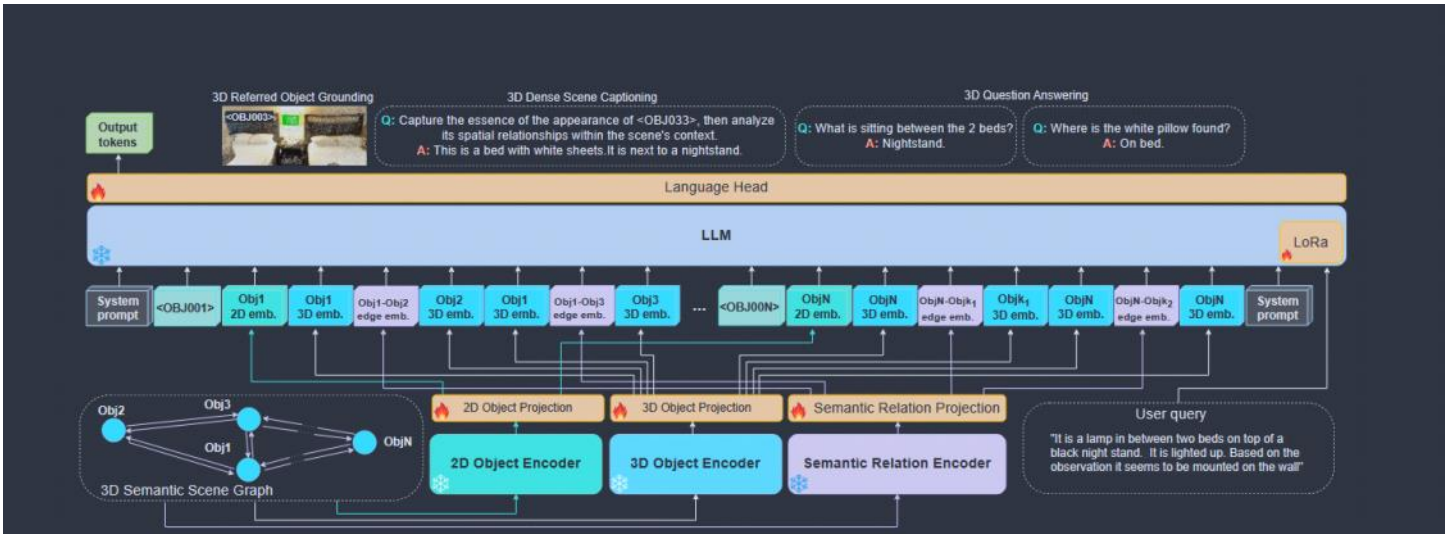


Figure 2. The overall architecture of our approach. We introduce trainable layers to map the extracted graph node and edge features into the token embedding space of a pre-trained LLM. The scene graph is flattened for input into the LLM, with each object represented by a subgraph of its k nearest neighbors. To further adapt the LLM to 3D vision-language tasks, we add new object tokens to the LLM's vocabulary alongside with objects' 2D features and fine-tune the LLM using LoRa.

KNN strategy for prompting

System:	A chat between a curious user and an artificial intelligence assistant. The assistant gives helpful, detailed, and polite answers to the user's questions. The conversation centers around an indoor scene: [<OBJ001> $F_1^{2d}, F_1^v, F_{12}^e, F_2^v F_1^v, F_{14}^e, F_4^v$ ...<OBJN> $F_N^{2d}, F_N^v, F_{Nk_1}^e, F_{k_1}^v F_N^v, F_{Nk_2}^e, F_{k_2}^v$]
User:	According to the given description, *there are brown wooden cabinets, placed on the side of the kitchen*, please provide the ID of the object that closely matches this description.
Assistant:	<OBJ001>.

Table 1. Example of prompt for the language model containing scene graph.

SG-Nav: Online 3D Scene Graph Prompting for LLM-based Zero-shot Object Navigation (NeurIPS 2024 / CVPR 2025 扩展版 UniGoal) - Hierarchical Chain-of-Thought Prompt + reperception mechanism

核心亮点：提出了一种在线构建分层 3D 场景图的零样本导航框架。它设计了**分层思维链（**Hierarchical Chain-of-Thought**）**，引导 LLM 通过遍历图的节点和边来推理目标位置。最惊艳的是它的“再感知（Re-perception）”机制，LLM

可以在导航过程中根据实时图节点持续判断目标的可信度，动态纠正 VLM 的感知幻觉。

Project Page:https://bagh2178.github.io/SG-Nav/

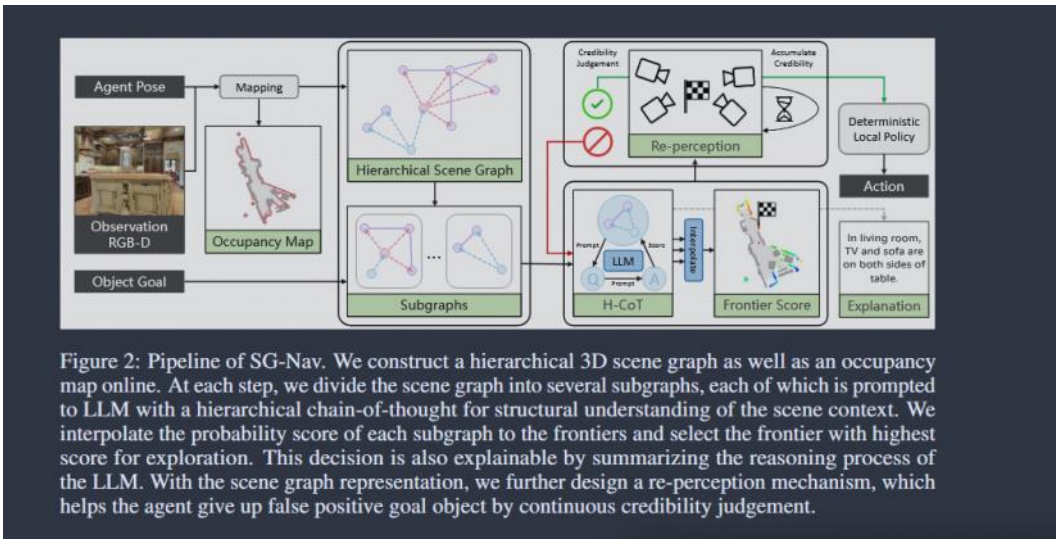


Figure 2: Pipeline of SG-Nav. We construct a hierarchical 3D scene graph as well as an occupancy map online. At each step, we divide the scene graph into several subgraphs, each of which is prompted to LLM with a hierarchical chain-of-thought for structural understanding of the scene context. We interpolate the probability score of each subgraph to the frontiers and select the frontier with highest score for exploration. This decision is also explainable by summarizing the reasoning process of the LLM. With the scene graph representation, we further design a re-perception mechanism, which helps the agent give up false positive goal object by continuous credibility judgement.

Setting：

Problem Definition. A mobile agent is tasked with navigating to an object o specified by its category c in an unknown environment. The agent receives streaming RGB-D video as input along with its

Hierarchical Scene Graph: Object-Group-Room + Incremental Updating and Pruning - pre-defined dictionary for group

6.2 Related Categories

When registering a new group node, we need to judge whether two object categories are related. We use LLM to generate the relationship dictionary, as shown below:

(Bed, Nightstand) (Wardrobe, Dresser) (Bookshelf, Chair) (Counter, Stove) (Table, Chair) (Bathroom Sink, Mirror) (Shower, Bathtub) (Refrigerator, Freezer) (Oven, Microwave) (Washing Machine, Dryer) (Sofa, Table) (Desk, Office Chair) (Computer, Monitor) (Piano, Bench) (Fireplace, Mantel) (Table, Mirror) (Window, Curtains) (Closet, Hangers) (Bathroom Cabinet, Toiletries) (Living Room Rug, Coffee Table) (Kitchen Cabinet, Dishes) (Dining Room Chandelier, Dining Table) (Clock, Wall) (Floor Lamp, Reading Chair) (Couch, Throw Pillows) (Bookcase, Books)

Cross-level notes generation

We also design different strategies to build edges between the nodes of different levels. We first connect high-level nodes with their affiliated low-level nodes. For room-object node pair, we think the object belongs to the room if the instance mask of this object on the point clouds is contained in the instance mask of this room. For room-group node pair, we connect them if all objects in this group belongs to the room. We do not additionally connect group and object nodes since the former is registered based on the latter.

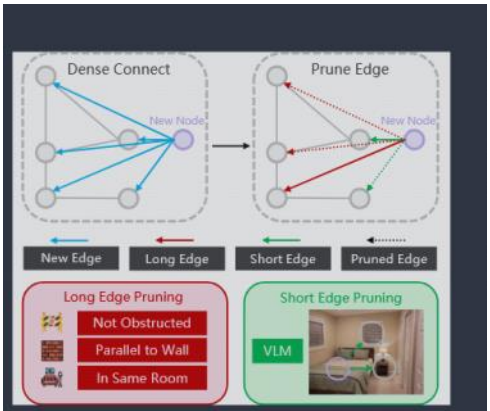


Figure 3: The incremental generation of edges. We densely connect newly registered nodes (purple) to all other nodes by efficiently prompting the LLM. We divide the edges into long edges and short edges and prune less informative ones with different strategies.

Prompting LLM with Scene Graph:

making. Since directly predicting a is hard for LLM, we adopt Frontier-based Exploration (FBE) strategy to divide the decision into two part: (1) synchronous with the 3D scene graph, we build an online occupancy map to indicate explored, unexplored and occupied area in BEV, based on which we compute the frontiers of explored area. In order to further explore the environment, the agent must go through one of the frontier. So in high level, the LLM is only required to select one of the frontier which has the highest probability to find o; (2) once the frontier to be explore is chosen, we simply apply local policy like Fast Marching Method [34] for path planning and output a.

预测子图中有目标概率：

Hierarchical Chain-of-thought Prompting. Different from previous methods [46; 52] that prompt LLM with only categories of objects near each frontier, we compute the probability P^{fro} of each frontier by prompting LLM with our 3D scene graph. At time t, we divide the scene graph $\mathcal{G}_t$ into subgraphs, each of which is determined by an object node with all its parent nodes and other directly connected object nodes. For each subgraph, we predict P^{sub}, the possibility of goal appearing near this subgraph. Then the probability P_i^{fro} of the i-th frontier can be averaged by:

$$P_i^{fro} = \sum_{j=1}^{M} \frac{P_j^{sub}}{D_{ij}} \quad (i \in \{1, 2, ..., N\}) \tag{2}$$

分层次Prompt

In order to make the prediction of P^{sub} accurate and explainable, we propose a hierarchical chain-of-thought method to prompt LLM, which fully exploits the structure information contained in the subgraphs. Specifically, for each subgraph, LLM is first prompted with *Predict the most likely distance between* [object] *and* [goal], where [object] means the category of the central object node. Then we prompt LLM to ask question related to the navigation goal, with the prompt: *Ask questions about the* [object] *and* [goal] *for predicting their distance*. After obtaining the questions about goal, we prompt LLM to answer those questions according to the information of subgraph, with the prompt: *Given* [nodes] *and* [edges] *of subgraph, answer above* [questions]. Therefore, from those answers we get all the necessary information required for inferring the position of goal. Finally, we prompt LLM to summarize the conversation and predict P, with the prompt *Based on the above conversation, determine the most likely distance between this subgraph and* [goal]. And P^{sub} can be simply computed by taking the inverse of the outputted distance. The LLM also summarizes the

Graph-based Re-perception

Graph-based Re-perception. Conventional object navigation framework does not take perception error into consideration. Once the agent detects out false positive objects of the goal category, it will not explore or perceive the environment any more and directly navigate to this object.

To solve this problem, we enhance the current object navigation framework with graph-based re-perception mechanism. When the agent detects out a goal object, it will approach to this object, observe it from multiple perspectives and accumulate a credibility score S. For the k-th RGB-D observation since this moment, we compute the credibility of it by:

$$S_k = C_k \cdot \sum_{j=1}^{M} \frac{P_j^{sub}}{D_j} \tag{3}$$

where C_k is the confidence score of this goal object, D_j is the distance between the central object node of the j-th subgraph and this object. The success condition for our re-perception mechanism is:

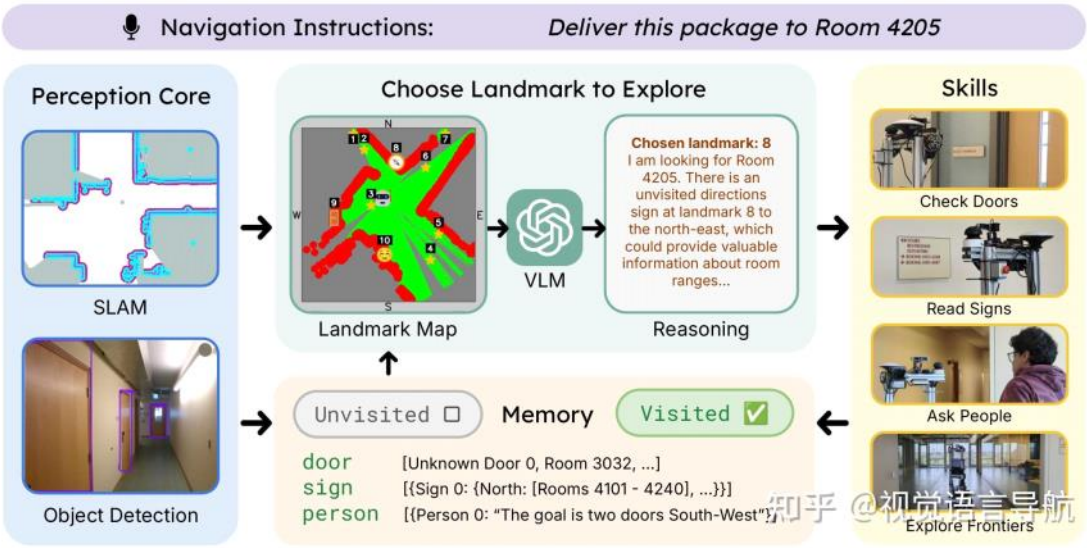
$$N_{stop} < N_{max}, \text{ where } N_{stop} = N, \text{ s.t. } \sum_{i=1}^{N-1} S_i < S_{thres} \leq \sum_{i=1}^{N} S_i \tag{4}$$

where N_{max} is a pre-defined maximum number of steps. If succeed, the agent will navigate to this goal object without any further exploration or perception. Otherwise, it will give up this object and continue the exploration.

7. ReasonNav - 通过阅读标识与询问他人方向辅助导航

- 论文链接：https://arxiv.org/pdf/2509.21189v1
- 项目主页：https://reasonnav.github.io/
- 代码链接：https://github.com/ReasonNav/Re

- 提出模块化的导航系统ReasonNav，通过利用视觉语言模型的推理能力，将人类在人造环境中导航时的行为（如阅读标识、询问他人方向等）整合到机器人导航中，使机器人能够更高效地在大型复杂建筑中导航。
- 设计了**基于导航地标**的紧凑输入和输出抽象，让VLM能够专注于语言理解和推理，从而更好地利用周围环境中的知识资源来提高导航效率。



地标的定义与作用

- 本方法的抽象设计以地标为中心，地标是指在导航任务中特别重要的显著物体。具体来说，地标包括三个类别的物体：门、人和方向标识，以及俯视图地图的边界。
- 系统从检测器的输出中填充地标的记忆库，并在执行各种技能时附加额外的导航相关信息。
- 例如，对于门，会附加与之关联的房间标签文本；对于人，会附加VLM生成的从他们那里收到的信息的摘要；对于方向标识，会附加每个方向上的标识文本以及相应的方向列表。
- 所有物体都会被标记为“已访问”或“未访问”。

VLM的输入和输出

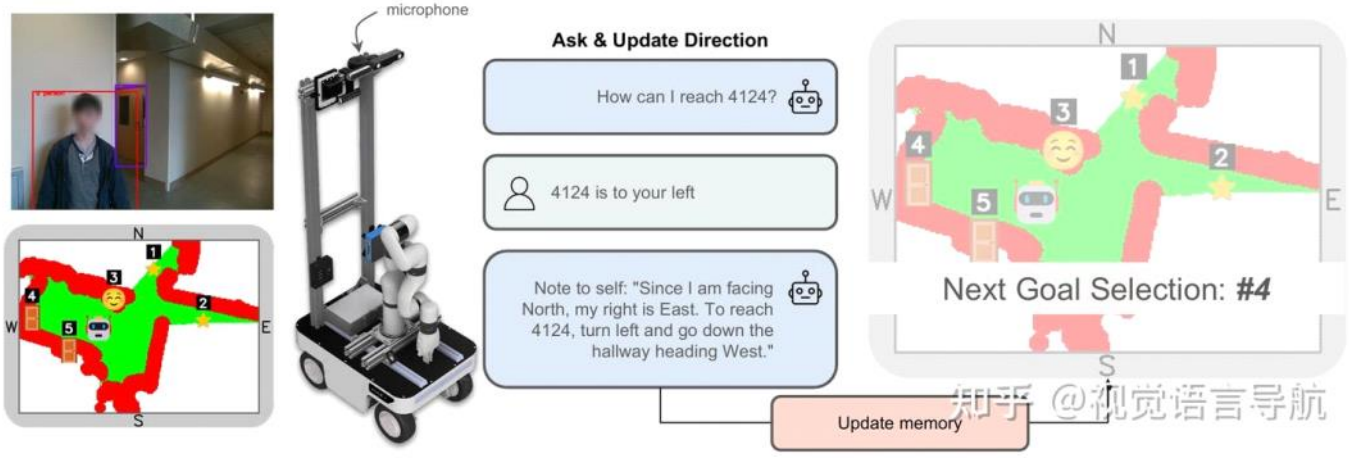
- VLM的输入包括文本指令和两种形式的抽象场景信息。
- 一种是包括物体和地图边界在内的地标记忆库，以JSON格式呈现。每个地标都被分配一个索引，并可能附加了上述提到的额外信息。
- 另一种信息是智能体当前的俯视图地图的图像可视化。地图根据占用情况和已探索区域进行着色，并为每个地标在其类别符号和索引号的位置上进行标记。
- VLM被提示使用这两种信息来决定下一步要访问的地标。这种设计确保了VLM可以灵活地选择任何合理的高级计划，而无需预测精确的数值坐标。具体的提示和详细的推理过程可以在附录中找到。

行为原语

- **边界（探索）**：智能体移动到期望的边界，并使用Nav2的点目标规划器和控制器进行360度旋转以扫描其周围环境。边界导航使我们能够探索未访问的区域并识别更多的地标。
- **门（房间标签读取）**：智能体接近门并平移其相机，同时调用目标检测器查询房间标签。如果检测到房间标签，智能

体会靠近并再次调用VLM来读取它。然后，将文本附加到记忆库中的门。如果找到了目标，任务结束。

- **人（询问方向）**：智能体接近人，并使用文本到语音模型询问方向。然后，它使用语音到文本记录人的回答。接下来，调用VLM生成一个关于接收到的信息的简短笔记，然后将其附加到记忆库中人的地标。重要的是，要求VLM在全局地图框架中使用基本方向，而不是相对方向（如“左”或“右”），以便在记录时不需要从记录时的智能体姿态理解笔记。



- **方向标识（标识读取）**：智能体接近标识并使用VLM读取它。根据箭头方向将标识文本分组（分为基本方向），并将方向转换为全局地图框架以便记录。